4 Thermodynamic Properties and Phase Equilibrium

A number of design calculations require a knowledge of thermodynamic properties and phase equilibrium. In practice, the designer most often uses a commercial physical property or a simulation software package to access such data. However, the designer must understand the basis of the methods for thermodynamic properties and phase equilibrium, so that the most appropriate methods can be chosen and their limitations fully understood.

4.1 EQUATIONS OF STATE

The relationship between pressure, volume and temperature for fluids is described by *equations of state*. For example, if a gas is initially at a specified pressure, volume and temperature and two of the three variables are changed, the third variable can be calculated from an equation of state.

Gases at low pressure tend towards *ideal gas* behavior. For a gas to be ideal,

- the volume of the molecules should be small compared with the total volume;
- there should be no intermolecular forces.

The behavior of ideal gases can be described by the ideal gas law[1,2]:

$$PV = NRT \tag{4.1}$$

where P = pressure ($\mathrm{N{\cdot}m^{-2}}$)
V = volume occupied by N kmol of gas ($\mathrm{m^3}$)
N = moles of gas (kmol)
R = gas constant (8314 $\mathrm{N{\cdot}m{\cdot}kmol^{-1}{\cdot}K^{-1}}$ or $\mathrm{J{\cdot}kmol^{-1}{\cdot}K^{-1}}$)
T = absolute temperature (K)

The ideal gas law describes the actual behavior of most gases reasonably well at pressures below 5 bar.

If standard conditions are specified to be 1 atm (101,325 $\mathrm{N{\cdot}m^{-2}}$) and 0°C (273.15 K), then from the ideal gas law, the volume occupied by 1 kmol of gas is 22.4 $\mathrm{m^3}$.

For gas mixtures, the *partial pressure* is defined as the pressure that would be exerted if that component alone occupied the volume of the mixture. Thus, for an ideal gas,

$$p_i V = N_i RT \tag{4.2}$$

where p_i = partial pressure ($\mathrm{N{\cdot}m^{-2}}$)
N_i = moles of Component i (kmol)

The mole fraction in the gas phase for an ideal gas is given by a combination of Equations 4.1 and 4.2:

$$y_i = \frac{N_i}{N} = \frac{p_i}{P} \tag{4.3}$$

where y_i = mole fraction of Component i

For a mixture of ideal gases, the sum of the partial pressures equals the total pressure (Dalton's Law):

$$\sum_i^{NC} p_i = P \tag{4.4}$$

where p_i = partial pressure of Component i ($\mathrm{N{\cdot}m^{-2}}$)
NC = number of components (–)

The behavior of real gases and liquids can be accounted for by introducing a compressibility factor (Z), such that[1–3]:

$$PV = ZRT \tag{4.5}$$

where Z = compressibility factor (–)
V = molar volume ($\mathrm{m^3{\cdot}kmol^{-1}}$)

$Z = 1$ for an ideal gas and is a function of temperature, pressure and composition for mixtures. A model for Z is needed.

In process design calculations, *cubic* equations of state are most commonly used. The most popular of these cubic equations is the Peng–Robinson equation of state given by[3]:

$$Z = \frac{V}{V-b} - \frac{aV}{RT(V^2 + 2bV - b^2)} \tag{4.6}$$

where

$$Z = \frac{PV}{RT} \tag{4.7}$$

V = molar volume

$$a = 0.45724\frac{R^2T_c^2}{P_c}\alpha$$

$$b = 0.0778\frac{RT_c}{P_c}$$

$$\alpha = (1 + \kappa(1 - \sqrt{T_R}))^2$$

$$\kappa = 0.37464 + 1.54226\,\omega - 0.26992\,\omega^2$$

T_C = critical temperature

P_C = critical pressure

ω = acentric factor

$$= \left[-\log\left(\frac{P^{SAT}}{P_C}\right)_{T_R=0.7}\right] - 1$$

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R = gas constant

$T_R = T/T_C$

T = absolute temperature

The acentric factor is obtained experimentally. It accounts for differences in molecular shape, increasing with non-sphericity and polarity, and tabulated values are available[3]. Equation 4.6 can be rearranged to give a cubic equation of the form:

$$Z^3 + \beta Z^2 + \gamma Z + \delta = 0 \tag{4.8}$$

where
$$Z = \frac{PV}{RT}$$
$$\beta = B - 1$$
$$\gamma = A - 3B^2 - 2B$$
$$\delta = B^3 + B^2 - AB$$
$$A = \frac{aP}{R^2T^2}$$
$$B = \frac{bP}{RT}$$

This is a cubic equation in the compressibility factor Z, which can be solved analytically. The solution of this cubic equation may yield either one or three real roots[4]. To obtain the roots, the following two values are first calculated[4]:

$$q = \frac{\beta^2 - 3\gamma}{9} \tag{4.9}$$

$$r = \frac{2\beta^3 - 9\beta\gamma + 27\delta}{54} \tag{4.10}$$

If $q^3 - r^2 \geq 0$, then the cubic equation has three roots. These roots are calculated by first calculating:

$$\theta = \arccos\left(\frac{r}{q^{3/2}}\right) \tag{4.11}$$

Then, the three roots are given by:

$$Z_1 = -2q^{1/2}\cos\left(\frac{\theta}{3}\right) - \frac{\beta}{3} \tag{4.12}$$

$$Z_2 = -2q^{1/2}\cos\left(\frac{\theta + 2\pi}{3}\right) - \frac{\beta}{3} \tag{4.13}$$

$$Z_3 = -2q^{1/2}\cos\left(\frac{\theta + 4\pi}{3}\right) - \frac{\beta}{3} \tag{4.14}$$

If $q^3 - r^2 < 0$, then the cubic equation has only one root given by:

$$Z_1 = -\text{sign}(r)\{[(r^2 - q^3)^{1/2} + |r|]^{1/3} + \frac{q}{[(r^2 - q^3)^{1/2} + |r|]^{1/3}}\} - \frac{\beta}{3} \tag{4.15}$$

where sign(r) is the sign of r, such that sign(r) = 1 if $r > 0$ and sign(r) = −1 if $r < 0$.

If there is only one root, there is no choice but to take this as the compressibility factor at the specified conditions of temperature and pressure, but if three real values exist, then the largest corresponds to the vapor compressibility factor and the smallest is the liquid compressibility factor. The middle value has no physical meaning. A superheated vapor might provide only one root, corresponding with the compressibility factor of the vapor phase. A subcooled liquid might provide only one root, corresponding with the compressibility factor of the liquid phase. A vapor–liquid system should provide three roots, with only the largest and smallest being significant. Equations of state such as the Peng–Robinson equation are generally more reliable at predicting the vapor compressibility than the liquid compressibility.

For multicomponent systems, mixing rules are needed to determine the values of a and b[3,5]:

$$b = \sum_i^{NC} x_i b_i \tag{4.16}$$

$$a = \sum_i^{NC}\sum_j^{NC} x_i x_j \sqrt{a_i a_j}(1 - k_{ij}) \tag{4.17}$$

where k_{ij} is a binary interaction parameter found by fitting experimental data[5].

Hydrocarbons	:	k_{ij} very small, nearly zero
Hydrocarbon/Light gas	:	k_{ij} small and constant
Polar mixtures	:	k_{ij} large and temperature dependent

Some interaction parameters have been published[5].

Example 4.1 Using the Peng–Robinson equation of state:

a. determine the vapor compressibility of nitrogen at 273.15 K and 1.013 bar, 5 bar and 50 bar, and compare with an ideal gas. For nitrogen, $T_C = 126.2$ K, $P_C = 33.98$ bar and $\omega = 0.037$. Take $R = 0.08314$ bar·m^3·kmol^{-1}·K^{-1}.
b. determine the liquid density of benzene at 293.15 K and compare this with the measured value of $\rho_L = 876.5$ kg·m^{-3}. For benzene, $T_C = 562.05$ K, $P_C = 48.95$ bar and $\omega = 0.210$.

Solution

a. For nitrogen, Equation 4.8 must be solved for the vapor compressibility factor. In this case, $q^3 - r^2 < 0$ and there is one root given by Equation 4.15. The solution is summarized in Table 4.1.
   From Table 4.1, it can be seen that the nitrogen can be approximated by ideal gas behavior at moderate pressures.
b. For benzene, at 293.15 K and 1.013 bar (1 atm), Equation 4.8 must be solved for the liquid compressibility factor. In

Table 4.1 Solution of the Peng–Robinson equation of state for nitrogen.

	Pressure 1.013 bar	Pressure 5 bar	Pressure 50 bar
κ	0.43133	0.43133	0.43133
α	0.63482	0.63482	0.63482
a	0.94040	0.94040	0.94040
b	2.4023×10^{-2}	2.4023×10^{-2}	2.4023×10^{-2}
A	1.8471×10^{-3}	9.1171×10^{-3}	9.1171×10^{-2}
B	1.0716×10^{-3}	5.2891×10^{-3}	5.2891×10^{-2}
β	−0.99893	0.99471	−0.94711
γ	-2.9947×10^{-4}	-1.5451×10^{-3}	-2.3004×10^{-2}
δ	-8.2984×10^{-7}	-2.0099×10^{-5}	-1.8767×10^{-3}
q	0.11097	0.11045	0.10734
r	-3.6968×10^{-2}	-3.6719×10^{-2}	-3.6035×10^{-2}
$q^3 - r^2$	-2.9848×10^{-8}	-7.1589×10^{-7}	-6.1901×10^{-5}
Z_1	0.9993	0.9963	0.9727
RT/P ($m^3 \cdot kmol^{-1}$)	22.42	4.542	0.4542
Z_1RT/P ($m^3 \cdot kmol^{-1}$)	22.40	4.525	0.4418

this case, $q^3 - r^2 > 0$ and there are three roots given by Equations 4.12 to 4.14. The parameters for the Peng–Robinson equation are given in Table 4.2.

Table 4.2 Solution of the Peng–Robinson equation of state for benzene.

	Pressure 1.013 bar
κ	0.68661
α	1.41786
a	28.920
b	7.4270×10^{-2}
A	4.9318×10^{-2}
B	3.0869×10^{-3}
β	−0.99691
γ	4.3116×10^{-2}
δ	-1.4268×10^{-4}
q	9.6054×10^{-2}
r	-2.9603×10^{-2}
$q^3 - r^2$	9.9193×10^{-6}
Z_1	0.0036094
Z_2	0.95177
Z_3	0.04153

From the three roots, only Z_1 and Z_2 are significant. The smallest root (Z_1) relates to the liquid and the largest root (Z_2) to the vapor. Thus, the density of liquid benzene is given by:

$$\rho_L = \frac{78.11}{Z_1 RT/P}$$
$$= \frac{78.11}{0.0036094 \times 24.0597}$$
$$= 899.5 \text{ kg m}^{-3}$$

This compares with an experimental value of $\rho_L = 876.5$ kg·m^{-3} (an error of 3%).

4.2 PHASE EQUILIBRIUM FOR SINGLE COMPONENTS

The phase equilibrium for pure components is illustrated in Figure 4.1. At low temperatures, the component forms a solid phase. At high temperatures and low pressures, the component forms a vapor phase. At high pressures and high temperatures, the component forms a liquid phase. The phase equilibrium boundaries between each of the phases are illustrated in Figure 4.1. The point where the three phase equilibrium boundaries meet is the *triple point*, where solid, liquid and vapor coexist. The phase equilibrium boundary between liquid and vapor terminates at the *critical point*. Above the critical temperature, no liquid forms, no matter how high the pressure. The phase equilibrium boundary between liquid and vapor connects the triple point and the

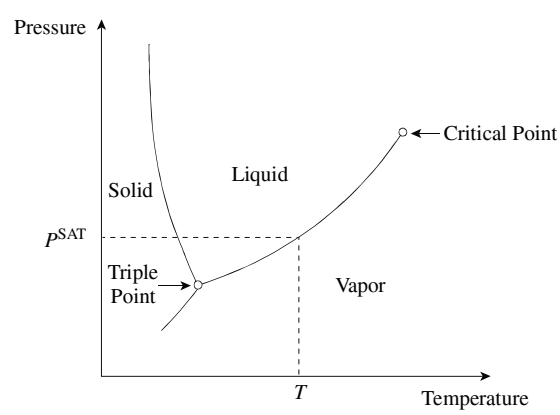


Figure 4.1 Phase equilibrium for a pure component.

critical point, and marks the boundary where vapor and liquid coexist. For a given temperature on this boundary, the pressure is the *vapor pressure*. When the vapor pressure is 1 atm, the corresponding temperature is the *normal boiling point*. If, at any given vapor pressure, the component is at a temperature less than the phase equilibrium, it is *subcooled*. If it is at a temperature above the phase equilibrium, it is *superheated*. Various expressions can be used to represent the vapor pressure curve.

The simplest expression is the Clausius–Clapeyron equation[1,2]:

$$\ln P^{SAT} = A - \frac{B}{T} \tag{4.18}$$

where A and B are constants and T is the absolute temperature. This indicates that a plot of $\ln P^{SAT}$ versus $1/T$ should be a straight line. Equation 4.18 gives a good correlation only over small temperature ranges. Various modifications have been suggested to extend the range of application, for example, the Antoine equation[1–3]:

$$\ln P^{SAT} = A - \frac{B}{C + T} \tag{4.19}$$

where A, B and C are constants determined by correlating experimental data[3]. Extended forms of the Antoine equation have also been proposed[3]. Great care must be taken when using correlated vapor pressure data not to use the correlation coefficients outside the temperature range over which the data has been correlated; otherwise, serious errors can occur.

4.3 FUGACITY AND PHASE EQUILIBRIUM

Having considered single component systems, multicomponent systems need to be addressed now. If a closed system contains more than one phase, the equilibrium condition can be written as:

$$f_i^I = f_i^{\mathrm{II}} = f_i^{III} \quad i = 1, 2, \ldots, NC \tag{4.20}$$

where f_i is the fugacity of Component i in Phases I, II and III and NC is the number of components. Fugacity is a thermodynamic pressure, but has no strict physical significance. It can be thought of as an "escaping tendency". Thus, Equation 4.20 states that if a system of different phases is in equilibrium, then the "escaping tendency" of Component i from the different phases is equal.

4.4 VAPOR–LIQUID EQUILIBRIUM

Thermodynamic equilibrium in a vapor–liquid mixture is given by the condition that the vapor and liquid fugacities for each component are equal[2]:

$$f_i^V = f_i^L \tag{4.21}$$

where f_i^V = fugacity of Component i in the vapor phase
f_i^L = fugacity of Component i in the liquid phase

Thus, equilibrium is achieved when the "escaping tendency" from the vapor and liquid phases for Component i are equal. The *vapor-phase fugacity coefficient*, ϕ_i^V, can be defined by the expression:

$$f_i^V = y_i \phi_i^V P \tag{4.22}$$

where y_i = mole fraction of Component i in the vapor phase
ϕ_i^V = vapor-phase fugacity coefficient
P = system pressure

The *liquid-phase fugacity coefficient* ϕ_i^L can be defined by the expression:

$$f_i^L = x_i \phi_i^L P \tag{4.23}$$

The *liquid-phase activity coefficient* γ_i can be defined by the expression:

$$f_i^L = x_i \gamma_i f_i^O \tag{4.24}$$

where x_i = mole fraction of Component i in the liquid phase
ϕ_i^L = liquid-phase fugacity coefficient
γ_i = liquid-phase activity coefficient
f_i^O = fugacity of Component i at standard state

For moderate pressures, f_i^O can be approximated by the saturated vapor pressure, P_i^{SAT}; thus, Equation 4.24 becomes[3]:

$$f_i^L = x_i \gamma_i P_i^{SAT} \tag{4.25}$$

Equations 4.21, 4.22 and 4.23 can be combined to give an expression for the K-value, K_i, that relates the vapor and liquid mole fractions:

$$K_i = \frac{y_i}{x_i} = \frac{\phi_i^L}{\phi_i^V} \tag{4.26}$$

Equation 4.26 defines the relationship between the vapor and liquid mole fractions and provides the basis for vapor–liquid equilibrium calculations on the basis of equations of state. Thermodynamic models are required for ϕ_i^V and ϕ_i^L from an equation of state. Alternatively, Equations 4.21, 4.22 and 4.25 can be combined to give

$$K_i = \frac{y_i}{x_i} = \frac{\gamma_i P_i^{SAT}}{\phi_i^V P} \tag{4.27}$$

This expression provides the basis for vapor–liquid equilibrium calculations on the basis of *liquid-phase activity coefficient models*. In Equation 4.27, thermodynamic models are required for ϕ_i^V (from an equation of state) and γ_i from a liquid-phase activity coefficient model. Some examples will be given later. At moderate pressures, the vapor phase becomes ideal, as discussed previously, and $\phi_i^V = 1$. For

an ideal vapor phase, Equation 4.27 simplifies to:

$$K_i = \frac{y_i}{x_i} = \frac{\gamma_i P_i^{SAT}}{P} \tag{4.28}$$

When the liquid phase behaves as an ideal solution,

- all molecules have the same size;
- all intermolecular forces are equal;
- the properties of the mixture depend only on the properties of the pure components comprising the mixture.

Mixtures of isomers, such as *o*-, *m*- and *p*-xylene mixtures, and adjacent members of homologous series, such as *n*-hexane–*n*-heptane and benzene–toluene mixtures, give close to ideal liquid-phase behavior. For this case, $\gamma_i = 1$, and Equation 4.28 simplifies to:

$$K_i = \frac{y_i}{x_i} = \frac{P_i^{SAT}}{P} \tag{4.29}$$

which is Raoult's law and represents both ideal vapor- and liquid-phase behavior. Correlations are available to relate component vapor pressure to temperature, as discussed above.

Comparing Equations 4.28 and 4.29, the liquid-phase nonideality is characterized by the activity coefficient γ_i. When $\gamma_i = 1$, the behavior is ideal. If $\gamma_i \neq 1$, then the value of γ_i can be used to characterize the nonideality:

- $\gamma_i < 1$ represents negative deviations from Raoult's Law;
- $\gamma_i > 1$ represents positive deviations from Raoult's Law.

The vapor–liquid equilibrium for noncondensable gases in equilibrium with liquids can often be approximated by Henry's Law[1–3]:

$$p_i = H_i x_i \tag{4.30}$$

where p_i = partial pressure of Component i
H_i = Henry's Law constant (determined experimentally)
x_i = mole fraction of Component i in the liquid phase

Assuming ideal gas behavior ($p_i = y_i P$):

$$y_i = \frac{H_i x_i}{P} \tag{4.31}$$

Thus, the K-value is given by:

$$K_i = \frac{y_i}{x_i} = \frac{H_i}{P} \tag{4.32}$$

A straight line would be expected from a plot of y_i against x_i.

The ratio of equilibrium K-values for two components measures their *relative volatility*:

$$\alpha_{ij} = \frac{K_i}{K_j} \tag{4.33}$$

where α_{ij} = volatility of Component i relative to Component j

These expressions form the basis for two alternative approaches to vapor–liquid equilibrium calculations:

a. $K_i = \phi_i^L / \phi_i^V$ forms the basis for calculations based entirely on equations of state. Using an equation of state for both the liquid and vapor phase has a number of advantages. Firstly, f_i^O need not be specified. Also, in principle, continuity at the critical point can be guaranteed with all thermodynamic properties derived from the same model. The presence of noncondensable gases, in principle, causes no additional complications. However, the application of equations of state is largely restricted to nonpolar components.
b. $K_i = \gamma_i P_i^{SAT} / \phi_i^V P$ forms the basis for calculations based on liquid-phase activity coefficient models. It is used when polar molecules are present. For most systems at low pressures, ϕ_i^V can be assumed to be unity. If high pressures are involved, then ϕ_i^V must be calculated from an equation of state. However, care should be taken when mixing and matching different models for γ_i and ϕ_i^V for high-pressure systems to ensure that appropriate combinations are taken.

Example 4.2 A gas from a combustion process has a flowrate of 10 $m^3 \cdot s^{-1}$ and contains 200 ppmv of oxides of nitrogen, expressed as nitric oxide (NO) at standard conditions of 0°C and 1 atm. This concentration needs to be reduced to 50 ppmv (expressed at standard conditions) before being discharged to the environment. It can be assumed that all of the oxides of nitrogen are present in the form of NO. One option being considered to remove the NO is by absorption in water at 20°C and 1 atm. The solubility of the NO in water follows Henry's Law, with $H_{NO} = 2.6 \times 10^4$ atm at 20°C. The gas is to be contacted countercurrently with water such that the inlet gas contacts the outlet water. The concentration of the outlet water can be assumed to reach 90% of equilibrium. Estimate the flowrate of water required, assuming the molar mass of gas in kilograms occupies 22.4 m^3 at standard conditions of 0°C and 1 atm.

Solution Molar flowrate of gas

$$= 10 \times \frac{1}{22.4}$$
$$= 0.446 \text{ kmol} \cdot \text{s}^{-1}$$

Assuming the molar flowrate of gas remains constant, the amount of NO to be removed

$$= 0.446(200 - 50) \times 10^{-6}$$
$$= 6.69 \times 10^{-5} \text{ kmol} \cdot \text{s}^{-1}$$

Assuming the water achieves 90% of equilibrium and contacts countercurrently with the gas, from

Henry's Law (Equation 4.38):

$$x_{NO} = \frac{0.9 y_{NO}^* P}{H_{NO}}$$

where y^*_{NO} = equilibrium mole fraction in the gas phase

$$x_{NO} = \frac{0.9 \times 200 \times 10^{-6} \times 1}{2.6 \times 10^4}$$
$$= 6.9 \times 10^{-9}$$

Assuming the liquid flowrate is constant, water flowrate

$$= \frac{6.69 \times 10^{-5}}{6.9 \times 10^{-9} - 0}$$
$$= 9696 \text{ kmol·s}^{-1}$$
$$= 9696 \times 18 \text{ kg·s}^{-1}$$
$$= 174{,}500 \text{ kg·s}^{-1}$$

This is an impractically large flowrate.

4.5 VAPOR–LIQUID EQUILIBRIUM BASED ON ACTIVITY COEFFICIENT MODELS

In order to model liquid-phase nonideality at moderate pressures, the liquid activity coefficient γ_i must be known:

$$K_i = \frac{\gamma_i P_i^{SAT}}{P} \tag{4.34}$$

γ_i varies with composition and temperature. There are three popular activity coefficient models[3],

a. Wilson
b. Nonrandom two liquid (NRTL)
c. Universal quasi-chemical (UNIQUAC)

These models are semiempirical and are based on the concept that intermolecular forces will cause nonrandom arrangement of molecules in the mixture. The models account for the arrangement of molecules of different sizes and the preferred orientation of molecules. In each case, the models are fitted to experimental binary vapor–liquid equilibrium data. This gives binary interaction parameters that can be used to predict multicomponent vapor–liquid equilibrium. In the case of the UNIQUAC equation, if experimentally determined vapor–liquid equilibrium data are not available, the Universal Quasi-chemical Functional Group Activity Coefficients (UNIFAC) method can be used to estimate UNIQUAC parameters from the molecular structures of the components in the mixture[3].

a. *Wilson equation*. The Wilson equation activity coefficient model is given by[3]:

$$\ln \gamma_i = -\ln\left[\sum_j^{NC} x_j \Lambda_{ij}\right] + 1 - \sum_k^{NC}\left[\frac{x_k \Lambda_{ki}}{\sum_j^N x_j \Lambda_{kj}}\right] \tag{4.35}$$

where

$$\Lambda_{ij} = \frac{V_j^L}{V_i^L}\exp\left[-\frac{\lambda_{ij} - \lambda_{ii}}{RT}\right] \tag{4.36}$$

where V_i^L = molar volume of pure Liquid i
λ_{ij} = energy parameter characterizing the interaction of Molecule i with Molecule j
R = gas constant
T = absolute temperature
$\Lambda_{ii} = \Lambda_{jj} = \Lambda_{kk} = 1$
$\Lambda_{ij} = 1$ for an ideal solution
$\Lambda_{ij} < 1$ for positive deviation from Raoult's Law
$\Lambda_{ij} > 1$ for negative deviation from Raoult's Law

For each binary pair, there are two adjustable parameters that must be determined from experimental data, that is, $(\lambda_{ij} - \lambda_{ii})$, which are temperature dependent. The ratio of molar volumes V_j^L/V_i^L is a weak function of temperature. For a binary system, the Wilson equation reduces to[3]:

$$\ln \gamma_1 = -\ln[x_1 + x_2\Lambda_{12}] + x_2\left[\frac{\Lambda_{12}}{x_1 + \Lambda_{12}x_2} - \frac{\Lambda_{21}}{x_1\Lambda_{21} + x_2}\right] \tag{4.37}$$

$$\ln \gamma_2 = -\ln[x_2 + x_1\Lambda_{21}] - x_1\left[\frac{\Lambda_{12}}{x_1 + \Lambda_{12}x_2} - \frac{\Lambda_{21}}{x_1\Lambda_{21} + x_2}\right] \tag{4.38}$$

where

$$\Lambda_{12} = \frac{V_2^L}{V_1^L}\exp\left[-\frac{\lambda_{12} - \lambda_{11}}{RT}\right]$$
$$\Lambda_{21} = \frac{V_1^L}{V_2^L}\exp\left[-\frac{\lambda_{21} - \lambda_{22}}{RT}\right] \tag{4.39}$$

The two adjustable parameters, $(\lambda_{12} - \lambda_{11})$ and $(\lambda_{21} - \lambda_{22})$, must be determined experimentally[6].

b. *NRTL equation*. The NRTL equation is given by[3]:

$$\ln \gamma_i = \frac{\sum_j^{NC} \tau_{ji} G_{ji} x_j}{\sum_k^{NC} G_{ki} x_k} + \sum_j^{NC} \frac{x_j G_{ij}}{\sum_k^{NC} G_{kj} x_k} \times \left(\tau_{ij} - \frac{\sum_k^{NC} x_k \tau_{kj} G_{kj}}{\sum_k^{NC} G_{kj} x_k}\right) \tag{4.40}$$

where $G_{ij} = \exp(-\alpha_{ij}\tau_{ij})$, $G_{ji} = \exp(-\alpha_{ij}\tau_{ji})$

$$\tau_{ij} = \frac{g_{ij} - g_{jj}}{RT}, \tau_{ji} = \frac{g_{ji} - g_{ii}}{RT}$$
$$G_{ij} \neq G_{ji}, \tau_{ij} \neq \tau_{ji}, G_{ii} = G_{jj} = 1, \tau_{ii} = \tau_{jj} = 0$$
$$\tau_{ij} = 0 \text{ for ideal solutions}$$

g_{ij} and g_{ji} are the energies of interactions between Molecules i and j. α_{ij} characterizes the tendency of Molecule i and Molecule j to be distributed in a random

fashion, depends on molecular properties and usually lies in the range 0.2 to 0.5. For each binary pair of components, there are three adjustable parameters, $(g_{ij} - g_{jj})$, $(g_{ji} - g_{ii})$ and $\alpha_{ij} (= \alpha_{ji})$, which are temperature dependent. For a binary system, the NRTL equation reduces to[3]:

$$\ln \gamma_1 = x_2^2 \left[\tau_{21} \left(\frac{G_{21}}{x_1 + x_2 G_{21}} \right)^2 + \frac{\tau_{12} G_{12}}{(x_2 + x_1 G_{12})^2} \right] \tag{4.41}$$

$$\ln \gamma_2 = x_1^2 \left[\tau_{12} \left(\frac{G_{12}}{x_2 + x_1 G_{12}} \right)^2 + \frac{\tau_{21} G_{21}}{(x_1 + x_2 G_{21})^2} \right] \tag{4.42}$$

where $G_{12} = \exp(-\alpha_{12}\tau_{12})$, $G_{21} = \exp(-\alpha_{21}\tau_{21})$,

$$\tau_{12} = \frac{g_{12} - g_{22}}{RT}, \qquad \tau_{21} = \frac{g_{21} - g_{11}}{RT}$$

The three adjustable parameters, $(g_{12} - g_{22})$, $(g_{21} - g_{11})$ and $\alpha_{12} (= \alpha_{21})$, must be determined experimentally[6].

c. *UNIQUAC equation.* The UNIQUAC equation is given by[3]:

$$\ln \gamma_i = \ln \left(\frac{\Phi_i}{x_i} \right) + \frac{z}{2} q_i \ln \left(\frac{\theta_i}{\Phi_i} \right) + l_i - \frac{\Phi_i}{x_i} \sum_j^{NC} x_j l_j + q_i \left[1 - \ln \left(\sum_j^{NC} \theta_j \tau_{ji} \right) - \sum_j^{NC} \frac{\theta_j \tau_{ij}}{\sum_k^{NC} \theta_k \tau_{kj}} \right] \tag{4.43}$$

where $\Phi_i = \dfrac{r_i x_i}{\sum_k^{NC} r_k x_k}$, $\theta_i = \dfrac{q_i x_i}{\sum_k^{NC} q_k x_k}$

$l_i = \dfrac{z}{2}(r_i - q_i) - (r_i - 1)$,

$\tau_{ij} = \exp - \left(\dfrac{u_{ij} - u_{jj}}{RT} \right)$

u_{ij} = interaction parameter between Molecule i and Molecule j $(u_{ij} = u_{ji})$

z = coordination number $(z = 10)$

r_i = pure component property, measuring the molecular van der Waals volume for Molecule i

q_i = pure component property, measuring the molecular van der Waals surface area for Molecule i

R = gas constant

T = absolute temperature

$u_{ij} = u_{ji}, \tau_{ii} = \tau_{jj} = 1$

For each binary pair, there are two adjustable parameters that must be determined from experimental data, that is, $(u_{ij} - u_{jj})$, which are temperature dependent. Pure component properties r_i and q_i measure molecular van der Waals volumes and surface areas and have been tabulated[6].

For a binary system, the UNIQUAC equation reduces to[3]:

$$\ln \gamma_1 = \ln \left(\frac{\Phi_1}{x_1} \right) + \frac{z}{2} q_1 \ln \left(\frac{\theta_1}{\Phi_1} \right) + \Phi_2 \left(l_1 - l_2 \frac{r_1}{r_2} \right) - q_1 \ln(\theta_1 + \theta_2 \tau_{21}) + \theta_2 q_1 \left(\frac{\tau_{21}}{\theta_1 + \theta_2 \tau_{21}} - \frac{\tau_{21}}{\theta_2 + \theta_1 \tau_{12}} \right) \tag{4.44}$$

$$\ln \gamma_2 = \ln \left(\frac{\Phi_2}{x_2} \right) + \frac{z}{2} q_2 \ln \left(\frac{\theta_2}{\Phi_2} \right) + \Phi_1 \left(l_2 - l_1 \frac{r_2}{r_1} \right) - q_2 \ln(\theta_2 + \theta_1 \tau_{12}) + \theta_1 q_2 \left(\frac{\tau_{12}}{\theta_2 + \theta_1 \tau_{12}} - \frac{\tau_{21}}{\theta_1 + \theta_2 \tau_{21}} \right) \tag{4.45}$$

where $\Phi_1 = \dfrac{r_1 x_1}{r_1 x_1 + r_2 x_2}$, $\Phi_2 = \dfrac{r_2 x_2}{r_1 x_1 + r_2 x_2}$

$\theta_1 = \dfrac{q_1 x_1}{q_1 x_1 + q_2 x_2}$, $\theta_2 = \dfrac{q_2 x_2}{q_1 x_1 + q_2 x_2}$

$l_1 = \dfrac{z}{2}(r_1 - q_1) - (r_1 - 1)$,

$l_2 = \dfrac{z}{2}(r_2 - q_2) - (r_2 - 1)$

$\tau_{12} = \exp - \left(\dfrac{u_{12} - u_{22}}{RT} \right)$,

$\tau_{21} = \exp - \left(\dfrac{u_{21} - u_{11}}{RT} \right)$

The two adjustable parameters, $(u_{12} - u_{22})$ and $(u_{21} - u_{11})$, must be determined experimentally[6]. Pure component properties r_1, r_2, q_1 and q_2 have been tabulated[6].

Since all experimental data for vapor–liquid equilibrium have some experimental uncertainty, it follows that the parameters obtained from data reduction are not unique[3]. There are many sets of parameters that can represent the experimental data equally well, within experimental uncertainty. The experimental data used in data reduction are not sufficient to fix a unique set of "best" parameters. Realistic data reduction can determine only a region of parameters[2].

Published interaction parameters are available[6]. However, when more than one set of parameters is available for a binary pair, which should be chosen?

a. Check if the experimental data is thermodynamically consistent. The Gibbs–Duhem equation[1,2] can be applied to experimental binary data to check its thermodynamic consistency and it should be consistent with this equation.
b. Choose parameters fitted at the process pressure.
c. Choose data sets covering the composition range of interest.
d. For multicomponent systems, choose parameters fitted to ternary or higher systems, if possible.

4.6 VAPOR–LIQUID EQUILIBRIUM BASED ON EQUATIONS OF STATE

Before an equation of state can be applied to calculate vapor–liquid equilibrium, the fugacity coefficient ϕ_i for each phase needs to be determined. The relationship between the fugacity coefficient and the volumetric properties can be written as:

$$\ln \phi_i = \frac{1}{RT} \int_V^\infty \left[\left(\frac{\partial P}{\partial N_i} \right)_{T,V,N_j} - \frac{RT}{V} \right] dV - RT \ln Z \tag{4.46}$$

For example, the Peng–Robinson equation of state for this integral yields[2]:

$$\ln \phi_i = \frac{b_i}{b}(Z-1) - \ln(Z-B) - \frac{A}{2\sqrt{2}B}\left(\frac{2\sum x_i a_i}{a} + \frac{b_i}{b} \right) \ln \left(\frac{Z + (1+\sqrt{2})B}{Z + (1-\sqrt{2})B} \right) \tag{4.47}$$

where $A = \dfrac{aP}{R^2T^2}$

$B = \dfrac{bP}{RT}$

Thus, given critical temperatures, critical pressures and acentric factors for each component, as well as a phase composition, temperature and pressure, the compressibility factor can be determined and hence component fugacity coefficients for each phase can be calculated. Taking the ratio of liquid to vapor fugacity coefficients for each component gives the vapor–liquid equilibrium K-value for that component. This approach has the advantage of consistency between the vapor- and liquid-phase thermodynamic models. Such models are widely used to predict vapor–liquid equilibrium for hydrocarbon mixtures and mixtures involving light gases.

A vapor–liquid system should provide three roots from the cubic equation of state, with only the largest and smallest being significant. The largest root corresponds to the vapor compressibility factor and the smallest is the liquid compressibility factor. However, some vapor–liquid mixtures can present problems. This is particularly so for mixtures involving light hydrocarbons with significant amounts of hydrogen, which are common in petroleum and petrochemical processes. Under some conditions, such mixtures can provide only one root for vapor–liquid systems, when there should be three. This means that both the vapor and liquid fugacity coefficients cannot be calculated and is a limitation of such cubic equations of state.

If an activity coefficient model is to be used at high pressure (Equation 4.27), then the vapor-phase fugacity coefficient can be predicted from Equation 4.47. However, this approach has the disadvantage that the thermodynamic models for the vapor and liquid phases are inconsistent. Despite this inconsistency, it might be necessary to use an activity coefficient model if there is reasonable liquid-phase nonideality, particularly with polar mixtures.

4.7 CALCULATION OF VAPOR–LIQUID EQUILIBRIUM

In the case of vapor–liquid equilibrium, the vapor and liquid fugacities are equal for all components at the same temperature and pressure, but how can this solution be found? In any phase equilibrium calculation, some of the conditions will be fixed. For example, the temperature, pressure and overall composition might be fixed. The task is to find values for the unknown conditions that satisfy the equilibrium relationships. However, this cannot be achieved directly. First, values of the unknown variables must be guessed and checked to see if the equilibrium relationships are satisfied. If not, then the estimates must be modified in the light of the discrepancy in the equilibrium, and iteration continued until the estimates of the unknown variables satisfy the requirements of equilibrium.

Consider a simple process in which a multicomponent feed is allowed to separate into a vapor and a liquid phase with the phases coming to equilibrium, as shown in Figure 4.2. An overall material balance and component material balances can be written as:

$$F = V + L \tag{4.48}$$

$$Fz_i = Vy_i + Lx_i \tag{4.49}$$

where F = feed flowrate (kmol·s^{-1})
V = vapor flowrate from the separator (kmol·s^{-1})
L = liquid flowrate from the separator (kmol·s^{-1})
z_i = mole fraction of Component i in the feed (–)
y_i = mole fraction of Component i in vapor (–)
x_i = mole fraction of Component i in liquid (–)

The vapor–liquid equilibrium relationship can be defined in terms of K-values by:

$$y_i = K_i x_i \tag{4.50}$$

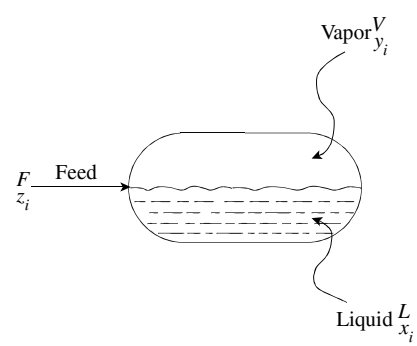


Figure 4.2 Vapor–liquid equilibrium.

Equations 4.48 to 4.50 can now be solved to give expressions for the vapor- and liquid-phase compositions leaving the separator:

$$y_i = \frac{z_i}{\dfrac{V}{F} + \left(1 - \dfrac{V}{F}\right)\dfrac{1}{K_i}} \tag{4.51}$$

$$x_i = \frac{z_i}{(K_i - 1)\dfrac{V}{F} + 1} \tag{4.52}$$

The vapor fraction (V/F) in Equations 4.51 and 4.52 lies in the range $0 \leq V/F \leq 1$.

For a specified temperature and pressure, Equations 4.51 and 4.52 need to be solved by trial and error. Given that:

$$\sum_i^{NC} y_i = \sum_i^{NC} x_i = 1 \tag{4.53}$$

where NC is the number of components, then:

$$\sum_i^{NC} y_i - \sum_i^{NC} x_i = 0 \tag{4.54}$$

Substituting Equations 4.51 and 4.52 into Equation 4.54, after rearrangement gives[7]:

$$\sum_i^{NC} \frac{z_i(K_i - 1)}{\dfrac{V}{F}(K_i - 1) + 1} = 0 = f(V/F) \tag{4.55}$$

To solve Equation 4.55, start by assuming a value of V/F and calculate $f(V/F)$ and search for a value of V/F until the function equals zero.

Many variations are possible around the basic *flash* calculation. Pressure and V/F can be specified and T calculated, and so on. Details can be found in King[7]. However, two special cases are of particular interest. If it is necessary to calculate the *bubble point*, then $V/F = 0$ in Equation 4.55, which simplifies to:

$$\sum_i^{NC} z_i(K_i - 1) = 0 \tag{4.56}$$

and given:

$$\sum_i^{NC} z_i = 1 \tag{4.57}$$

This simplifies to the expression for the bubble point:

$$\sum_i^{NC} z_i K_i = 1 \tag{4.58}$$

Thus, to calculate the bubble point for a given mixture and at a specified pressure, a search is made for a temperature to satisfy Equation 4.58. Alternatively, temperature can be specified and a search made for a pressure, the *bubble pressure*, to satisfy Equation 4.58.

Another special case is when it is necessary to calculate the *dew point*. In this case, $V/F = 1$ in Equation 4.55, which simplifies to:

$$\sum_i^{NC} \frac{z_i}{K_i} = 1 \tag{4.59}$$

Again, for a given mixture and pressure, temperature is searched to satisfy Equation 4.59. Alternatively, temperature is specified and pressure searched for the *dew pressure*.

If the K-value requires the composition of both phases to be known, then this introduces additional complications into the calculations. For example, suppose a bubble-point calculation is to be performed on a liquid of known composition using an equation of state for the vapor–liquid equilibrium. To start the calculation, a temperature is assumed. Then, calculation of K-values requires knowledge of the vapor composition to calculate the vapor-phase fugacity coefficient, and that of the liquid composition to calculate the liquid-phase fugacity coefficient. While the liquid composition is known, the vapor composition is unknown and an initial estimate is required for the calculation to proceed. Once the K-value has been estimated from an initial estimate of the vapor composition, the composition of the vapor can be reestimated, and so on.

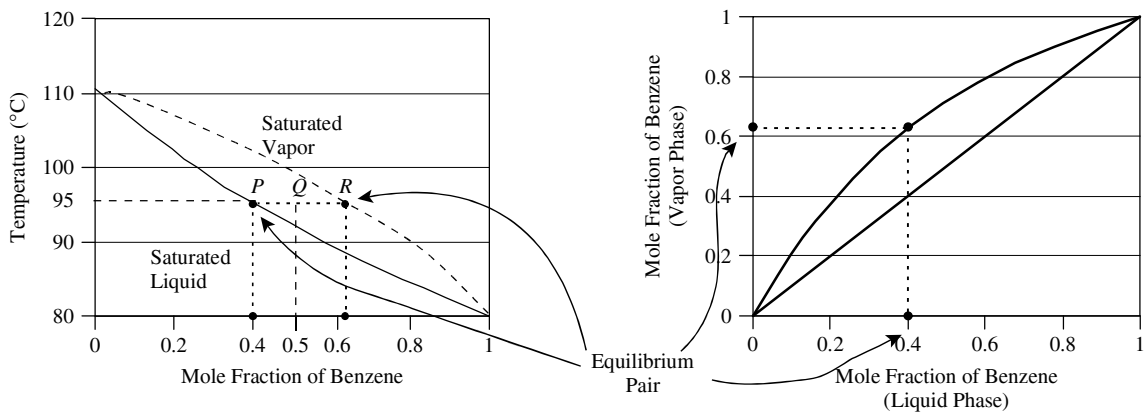
Figure 4.3 shows, as an example, the vapor–liquid equilibrium behavior for a binary mixture of benzene and toluene[8]. Figure 4.3a shows the behavior of temperature of the saturated liquid and saturated vapor (i.e. equilibrium pairs) as the mole fraction of benzene is varied (the balance being toluene). This can be constructed by calculating the bubble and dew points for different concentrations. Figure 4.3b shows an alternative way of representing the vapor–liquid equilibrium in a composition or *x–y diagram*. The x–y diagram can be constructed from the relative volatility. From the definition of relative volatility for a binary mixture of Components A and B:

$$\alpha_{AB} = \frac{y_A/x_A}{y_B/x_B} = \frac{y_A/x_A}{(1 - y_A)/(1 - x_A)} \tag{4.60}$$

Rearranging gives:

$$y_A = \frac{x_A \alpha_{AB}}{1 + x_A(\alpha_{AB} - 1)} \tag{4.61}$$

Thus, by knowing α_{AB} from vapor–liquid equilibrium and by specifying x_A, y_A can be calculated. Figure 4.3a also shows a typical vapor–liquid equilibrium pair, where the mole fraction of benzene in the liquid phase is 0.4 and that in the vapor phase is 0.62. A diagonal line across the x–y diagram represents equal vapor and liquid compositions. The phase equilibrium behavior shows a curve above the diagonal line. This indicates that benzene has a higher concentration in the vapor phase than toluene, that is,



(a) Temperature–composition behaviour.

(b) x–y representation.

Figure 4.3 Vapor–liquid equilibrium for a binary mixture of benzene and toluene at a pressure of 1 atm. (From Smith R and Jobson M, 2000, Distillation, Encyclopedia of Separation Science, Academic Press; reproduced by permission).

benzene is the more volatile component. Figure 4.3b shows the same vapor–liquid equilibrium pair as that shown in Figure 4.3a[8].

Figure 4.3a can be used to predict the separation in a single equilibrium stage, given a specified feed to the stage and a stage temperature. For example, suppose the feed is a mixture with equal mole fractions of benzene and toluene of 0.5 and this is brought to equilibrium at 95°C (Point Q in Figure 4.3a). Then, the resulting liquid will have a mole fraction of benzene of 0.4 and the vapor, a mole fraction of 0.62. In addition, the quantity of each phase formed can be determined from the lengths of the lines PQ and QR in Figure 4.3a. An overall material balance across the separator gives:

$$m_Q = m_P + m_R \tag{4.62}$$

A material balance for Component i gives:

$$m_Q x_{i,Q} = m_p x_{i,P} + m_R x_{i,R} \tag{4.63}$$

Substituting Equation 4.62 into Equation 4.63 and rearranging gives:

$$\frac{m_p}{m_R} = \frac{x_{i,R} - x_{i,Q}}{x_{i,Q} - x_{i,P}} \tag{4.64}$$

Thus, in Figure 4.3:

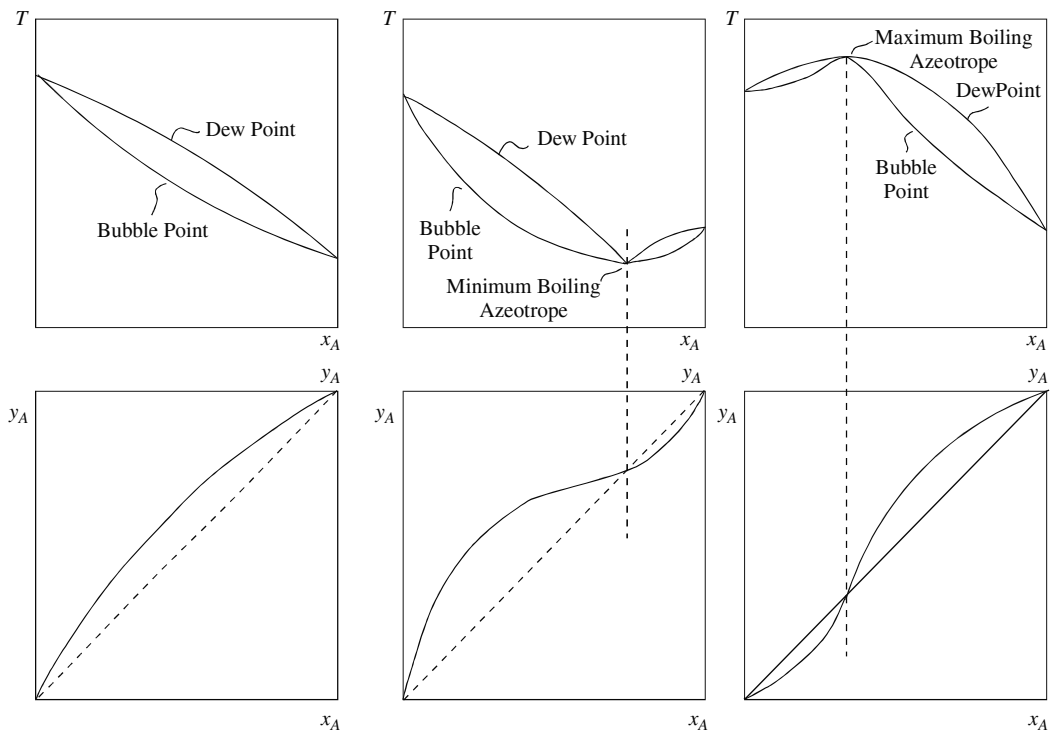
$$\frac{m_p}{m_R} = \frac{PQ}{QR} \tag{4.65}$$

The ratio of molar flowrates of the vapor and liquid phases is thus given by the ratio of the opposite line segments. This is known as the *Lever Rule,* after the analogy with a lever and fulcrum[7].

Consider first, a binary mixture of two Components A and B; the vapor–liquid equilibrium exhibits only a moderate deviation from ideality, as represented in Figure 4.4a. In this case, as pure A boils at a lower temperature than pure B in the temperature–composition diagram in Figure 4.4a, Component A is more volatile than Component B. This is also evident from the vapor–liquid composition diagram (x–y diagram), as it is above the line of $y_A = x_A$. In addition, it is also clear from Figure 4.4a that the order of volatility does not change as the composition changes. By contrast, Figure 4.4b shows a more highly nonideal behavior in which $\gamma_i > 1$ (positive deviation from Raoult's Law) forms a *minimum-boiling azeotrope*. At the azeotropic composition, the vapor and liquid are both at the same composition for the mixture. The lowest boiling temperature is below that of either of the pure components and is at the minimum-boiling azeotrope. It is clear from Figure 4.4b that the order of volatility of Components A and B changes, depending on the composition. Figure 4.4c also shows azeotropic behavior. This time, the mixture shows a behavior in which $\gamma_i < 1$ (negative deviation from Raoult's Law) forms a *maximum-boiling azeotrope*. This maximum-boiling azeotrope boils at a higher temperature than either of the pure components and would be the last fraction to be distilled, rather than the least volatile component, which would be the case with nonazeotropic behavior. Again, from Figure 4.4c, it can be observed that the order of volatility of Components A and B changes depending on the composition. Minimum-boiling azeotropes are much more common than maximum-boiling azeotropes.

Some general guidelines for vapor–liquid mixtures in terms of their nonideality are:

a. Mixtures of isomers usually form ideal solutions.
b. Mixtures of close-boiling aliphatic hydrocarbons are nearly ideal below 10 bar.
c. Mixtures of compounds close in molar mass and structure frequently do not deviate greatly from ideality (e.g. ring compounds, unsaturated compounds, naphthenes etc.).
d. Mixtures of simple aliphatics with aromatic compounds deviate modestly from ideality



(a) Moderate deviations from ideality. (b) Minimum boiling azeotrope. (c) Maximum boiling azeotrope.

Figure 4.4 Binary vapor-liquid equilibrium behavior.

e. Noncondensables such as CO_2, H_2S, H_2, N_2, and so on, that are present in mixtures involving heavier components tend to behave nonideally with respect to the other compounds.
f. Mixtures of polar and nonpolar compounds are always strongly nonideal.
g. Azeotropes and phase separation into liquid–liquid mixtures represent the ultimate in nonideality.

Moving down the list, the nonideality of the system increases.

Example 4.3 A mixture of ethane, propane, *n*-butane, *n*-pentane and *n*-hexane is given in the Table 4.3. For this calculation, it can be assumed that the K-values are ideal. For the mixture in Table 4.3, an equation of state method might have been a more appropriate choice. However, this makes the calculation of the K-values much more complex. The ideal K-values for the mixture can be expressed in terms of the Antoine Equation as:

$$K_i = \frac{1}{P}\exp\left[A_i - \frac{B_i}{T + C_i}\right] \tag{4.66}$$

where P is the pressure (bar), T the absolute temperature (K) and A_i, B_i and C_i are constants given in the Table 4.3:

a. For a pressure of 5 bar, calculate the bubble point.
b. For a pressure of 5 bar, calculate the dew point.
c. Calculate the pressure needed for total condensation at 313 K.
d. At a pressure of 6 bar and a temperature of 313 K, how much liquid will be condensed?

Table 4.3 Feed and physical property for components.

Component	Formula	Feed (kmol)	A_i	B_i	C_i
1. Ethane	C_2H_6	5	9.0435	1511.4	−17.16
2. Propane	C_3H_8	25	9.1058	1872.5	−25.16
3. *n*-Butane	C_4H_{10}	30	9.0580	2154.9	−34.42
4. *n*-Pentane	C_5H_{12}	20	9.2131	2477.1	−39.94
5. *n*-Hexane	C_6H_{14}	20	9.2164	2697.6	−48.78

Solution

a. The bubble point can be calculated from Equation 4.58 or from Equation 4.55 by specifying $V/F = 0$. The bubble point is calculated from Equation 4.58 in Table 4.4 to be 296.4 K. The search can be readily automated in spreadsheet software.

b. The dew point can be calculated from Equation 4.59 or 4.55 by specifying $V/F = 1$. In Table 4.5, the dew point is calculated from Equation 4.59 to be 359.1 K.

c. To calculate the pressure needed for total condensation at 313 K, the bubble pressure is calculated. The calculation is essentially the same as that in Part a above, with the temperature fixed at 313 K and the pressure varied, instead, in an iterative fashion until Equation 4.64 is satisfied. The resulting bubble pressure at 313 K is 7.4 bar.

If distillation was carried out giving an overhead composition, as given in Table 4.3, and if cooling water was used in the condenser, then total condensation at 313 K (40°C) would require an operating pressure of 7.4 bar.

Table 4.4 Bubble-point calculation.

i	z_i	$T = 275$ K		$T = 350$ K		$T = 300$ K		$T = 290$ K		$T = 296.366$ K	
		K_i	$z_i K_i$	K_i	$z_i K_i$	K_i	$z_i K_i$	K_i	$z_i K_i$	K_I	$z_i K_i$
1	0.05	4.8177	0.2409	18.050	0.9025	8.0882	0.4044	6.6496	0.3325	7.5448	0.3772
2	0.25	1.0016	0.2504	5.6519	1.4130	1.9804	0.4951	1.5312	0.3828	1.8076	0.4519
3	0.30	0.2212	0.0664	1.8593	0.5578	0.5141	0.1542	0.3742	0.1123	0.4594	0.1378
4	0.20	0.0532	0.0106	0.6802	0.1360	0.1464	0.0293	0.1000	0.0200	0.1279	0.0256
5	0.20	0.0133	0.0027	0.2596	0.0519	0.0437	0.0087	0.0280	0.0056	0.0373	0.0075
	1.00		0.5710		3.0612		1.0917		0.8532		1.0000

Table 4.5 Dew-point calculation.

i	z_i	$T = 325$ K		$T = 375$ K		$T = 350$ K		$T = 360$ K		$T = 359.105$ K	
		K_i	z_i/K_i	K_i	z_i/K_i	K_i	z_i/K_i	K_i	z_i/K_i	K_i	z_i/K_i
1	0.05	12.483	0.0040	24.789	0.0020	18.050	0.0028	20.606	0.0024	20.370	0.0025
2	0.25	3.4951	0.0715	8.5328	0.0293	5.6519	0.0442	6.7136	0.0372	6.6138	0.0378
3	0.30	1.0332	0.2903	3.0692	0.0977	1.8593	0.1614	2.2931	0.1308	2.2517	0.1332
4	0.20	0.3375	0.5925	1.2345	0.1620	0.6802	0.2941	0.8730	0.2291	0.8543	0.2341
5	0.20	0.1154	1.7328	0.5157	0.3879	0.2596	0.7704	0.3462	0.5778	0.3376	0.5924
	1.00		2.6911		0.6789		1.2729		0.9773		1.0000

Table 4.6 Flash calculation.

i	z_i	$V/F = 0$		$V/F = 0.5$	$V/F = 0.1$	$V/F = 0.05$	$V/F = 0.0951$
		K_i	$\frac{z_i(K_i-1)}{\frac{V}{F}(K_i-1)+1}$	$\frac{z_i(K_i-1)}{\frac{V}{F}(K_i-1)+1}$	$\frac{z_i(K_i-1)}{\frac{V}{F}(K_i-1)+1}$	$\frac{z_i(K_i-1)}{\frac{V}{F}(K_i-1)+1}$	$\frac{z_i(K_i-1)}{\frac{V}{F}(K_i-1)+1}$
1	0.05	8.5241	0.3762	0.0790	0.2147	0.2734	0.2193
2	0.25	2.2450	0.3112	0.1918	0.2768	0.2930	0.2783
3	0.30	0.6256	−0.1123	−0.1382	−0.1167	−0.1145	−0.1165
4	0.20	0.1920	−0.1616	−0.2711	−0.1758	−0.1684	−0.1751
5	0.20	0.0617	−0.1877	−0.3535	−0.2071	−0.1969	−0.2060
	1.00		0.2258	−0.4920	−0.0081	0.0866	0.0000

d. Equation 4.55 is used to determine how much liquid will be condensed at a pressure of 6 bar and 313 K. The calculation is detailed in Table 4.6, giving $V/F = 0.0951$, thus, 90.49% of the feed will be condensed.

Simple phase equilibrium calculations, like the one illustrated here, can be readily implemented in spreadsheet software and automated. In practice, the calculations will most often be carried out in commercial physical property packages, allowing more elaborate methods for calculating the equilibrium K-values to be used.

Example 4.4 Calculate the vapor composition of an equimolar liquid mixture of methanol and water at 1 atm (1.013 bar)

a. assuming ideal vapor- and liquid-phase behavior, that is, using Raoult's Law
b. using the Wilson equation.

Vapor pressure in bar can be predicted for temperature in Kelvin from the Antoine equation using coefficients in Table 4.7[3]. Data for the Wilson equation are given in Table 4.8[6]. Assume the gas constant $R = 8.3145$ kJ·kmol^{-1}·K^{-1}.

Table 4.7 Antoine coefficients for methanol and water[6].

	A_i	B_i	C_i
Methanol	11.9869	3643.32	−33.434
Water	11.9647	3984.93	−39.734

Table 4.8 Data for methanol (1) and water (2) for the Wilson equation at 1 atm[6].

V_1 (m^3·kmol^{-1})	V_2 (m^3·kmol^{-1})	$(\lambda_{12} - \lambda_{11})$ (kJ·kmol^{-1})	$(\lambda_{21} - \lambda_{22})$ (kJ·kmol^{-1})
0.04073	0.01807	347.4525	2179.8398

Table 4.10 Bubble-point calculation for a methanol–water mixture using the Wilson equation.

z_i	$T = 340$ K			$T = 350$ K			$T = 346.13$ K		
	γ_i	K_i	$z_i K_i$	γ_i	K_i	$z_i K_i$	γ_i	K_i	$z_i K_i$
0.5	1.1429	1.2501	0.6251	1.1363	1.8092	0.9046	1.1388	1.5727	0.7863
0.5	1.2307	0.3289	0.1645	1.2227	0.5012	0.2506	1.2258	0.4273	0.2136
1.00			0.7896			1.1552			0.9999

Solution

a. K-values assuming ideal vapor and liquid behavior are given by Raoult's Law (Equation 4.66). The composition of the liquid is specified to be $x_1 = 0.5$, $x_2 = 0.5$ and the pressure 1 atm, but the temperature is unknown. Therefore, a bubble-point calculation is required to determine the vapor composition. The procedure is the same as that in Example 4.2 and can be carried out in spreadsheet software. Table 4.9 shows the results for Raoult's Law.

Table 4.9 Bubble-point calculation for an ideal methanol–water mixture.

z_i	$T = 340$ K		$T = 360$ K		$T = 350$ K	
	K_i	$z_i K_i$	K_i	$z_i K_i$	K_i	$z_i K_i$
0.5	1.0938	0.5469	2.2649	1.1325	1.5906	0.7953
0.5	0.2673	0.1336	0.6122	0.3061	0.4094	0.2047
1.00		0.6805		1.4386		1.0000

Thus, the composition of the vapor at 1 atm is $y_1 = 0.7953$, $y_2 = 0.2047$, assuming an ideal mixture.

b. The activity coefficients for the methanol and water can be calculated using the Wilson equation (Equations 4.37 to 4.38). The results are summarized in Table 4.10.

Thus, the composition of the vapor phase at 1 atm is $y_1 = 0.7863$, $y_2 = 0.2136$ from the Wilson Equation. For this mixture, at these conditions, there is not much difference between the predictions of Raoult's Law and the Wilson equation, indicating only moderate deviations from ideality at the chosen conditions.

Example 4.5 2-Propanol (isopropanol) and water form an azeotropic mixture at a particular liquid composition that results in the vapor and liquid compositions being equal. Vapor–liquid equilibrium for 2-propanol–water mixtures can be predicted by the Wilson equation. Vapor pressure coefficients in bar with temperature in Kelvin for the Antoine equation are given in Table 4.11[3]. Data for the Wilson equation are given in Table 4.12[6]. Assume the gas constant $R = 8.3145$ kJ·kmol^{-1}·K^{-1}. Determine the azeotropic composition at 1 atm.

Table 4.11 Antoine equation coefficients for 2-propanol and water[6].

	A_i	B_i	C_i
2-propanol	13.8228	4628.96	−20.524
Water	11.9647	3984.93	−39.734

Table 4.12 Data for 2-propanol (1) and water (2) for the Wilson equation at 1 atm[6].

V_1 (m^3·kmol^{-1})	V_2 (m^3·kmol^{-1})	$(\lambda_{12} - \lambda_{11})$ (kJ·kmol^{-1})	$(\lambda_{21} - \lambda_{22})$ (kJ·kmol^{-1})
0.07692	0.01807	3716.4038	5163.0311

Solution To determine the location of the azeotrope for a specified pressure, the liquid composition has to be varied and a bubble-point calculation performed at each liquid composition until a composition is identified, whereby $x_i = y_i$. Alternatively, the vapor composition could be varied and a dew-point calculation performed at each vapor composition. Either way, this requires iteration. Figure 4.5 shows the x–y diagram for the 2-propanol–water system. This was obtained by carrying out a bubble-point calculation at different values of the liquid composition. The point where the x–y plot crosses the diagonal line gives the azeotropic composition. A more direct search for the azeotropic composition can be carried out for such a binary system in a spreadsheet by varying T and x_1 simultaneously and by solving the objective function (see Section 3.9):

$$(x_1 K_1 + x_2 K_2 - 1)^2 + (x_1 - x_1 K_1)^2 = 0$$

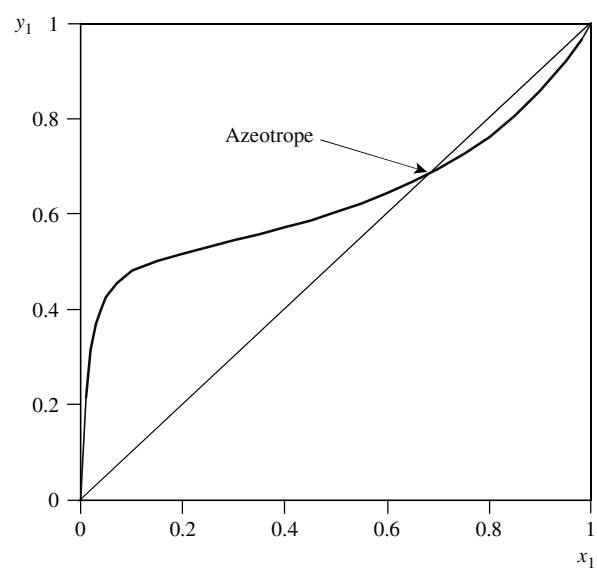


Figure 4.5 x–y plot for the system 2-propanol (1) and water (2) from the Wilson equation at 1 atm.